

Surname: \_\_\_\_\_

Initials: \_\_\_\_\_

St No. \_\_\_\_\_

(1) Print on A4 paper. (2) While you are preparing for the lab session, refer to the laboratory manual to complete the preparation sheet in your own handwriting in coloured ink → no black ink or ordinary pencil may be used. (3) If a heading is not applicable to your experiment, leave it open. (4) You have to leave the sheet at the completed experiment in your report book. (5) If you do not hand in your preparation, you will lose two marks. (6) You will earn either 1 or 2 **bonus marks** depending on the quality of your preparation. (7) Copying of someone else's work will be viewed in a serious light.

**1. Practical abbreviation and full name**

CLM: Conservation of Linear Momentum

**2. Goal**

To determine whether the law of conservation of momentum is adhered to between the two discs before and after the collision.

**3. Safety precautions**

Be careful when getting on and off the laboratory chair.

**4. Basic equations used in the practical**

$$V = \frac{\Delta S}{\Delta t} \quad p = m \cdot v$$

**5. Data to be collected**

Distances travelled of discs ~~time~~ to determine the speeds of the discs. Mass of the discs. Momentum of the discs.

**6. Table headings including Data collected and calculated data. Mention units of each measurement.**

|                   | Before  |        | After   |        |
|-------------------|---------|--------|---------|--------|
|                   | striker | target | striker | target |
| Distance (m)      |         |        |         |        |
| Time (s)          |         |        |         |        |
| Momentum (kg·m/s) |         |        |         |        |
| Angle (°)         |         |        |         |        |
| Mass (kg)         |         |        |         |        |

(P)

**7. Sketch the axes of the graph(s) to be drawn with axis labels and units. What type of graph paper will be used?**

Velocity (m/s)

N/A

**8. What data will be obtained from the graph? Include calculation formula (including units).**

N/A